

# Soham Desai

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## SKILLS

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**Languages:** Python, TypeScript, JavaScript, SQL

**Backend:** Flask, FastAPI, Django, Node.js, Express.js, REST APIs

**Frontend:** React, HTML, CSS, Tailwind CSS

**Databases & ORM:** PostgreSQL, MySQL, SQL Server, SQLAlchemy

**DevOps & Cloud:** Docker, AWS (EC2, S3, CloudWatch), CI/CD, Git/GitHub, Nginx

**Testing & Tools:** Pytest, Selenium, REST API Testing, Linux, Agile Development

## WORK EXPERIENCE

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### SIGNAI

Software Engineer Intern

Toronto, ON

May 2026 – August 2026

- Deployed and maintained a production Flask-based ASL recognition API using Docker and AWS EC2, supporting real-time letter and word recognition through machine-learning models.
- Implemented cloud-based model management using AWS S3, enabling the application to discover and load updated ML models without requiring manual model deployment to the server.
- Built infrastructure monitoring with AWS CloudWatch, tracking application health, prediction requests, inference performance, container resource usage, authentication events, and session activity.
- Containerized backend services with Docker, configuring Gunicorn, Redis, networking, environment variables, restart policies, and production deployment workflows.
- Implemented and tested REST API endpoints for prediction, word recognition, session management, health checks, analytics, and multi-model status monitoring.

### AI FINANCIAL

Data Analyst Co-op

Richmond Hill, ON

May 2025 – August 2025

- Built and deployed Python automation pipelines end-to-end that replaced manual workflows, reducing team data processing effort by 40% in a live production environment.
- Developed and optimized SQL Server stored procedures for high-volume systems, improving query reliability and execution time through root-cause analysis and systematic debugging with logs and tracing.
- Designed and maintained Power BI dashboards that surfaced raw data as actionable business insights for cross-functional teams, presenting results in regular team demos.
- Authored centralized technical documentation covering automation workflows, data schemas, and codebase processes, accelerating developer onboarding across the team.

## PROJECTS

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### AIRLINE TRAVEL ASSISTANT | React, Node.js, Selenium, REST APIs | [Github Repo](#)

- Shipped a production-ready full-stack application with a React front-end and an Express/Node.js back-end integrating third-party REST APIs, achieving 98% API reliability through input validation, error handling, and async processing.
- Designed fault-tolerant RESTful microservice endpoints and engineered Selenium test suites that caught 15+ defects pre-release; reduced defect backlog by 20% over two sprint cycles.

### SECURING FILE SHARING APPLICATION | Python, Django, SQLAlchemy | [Github Repo](#)

- Reduced unauthorized access risk to zero by implementing role-based access control (RBAC) with strict permission boundaries across all user roles
- Improved data retrieval performance by 35% by designing optimized PostgreSQL schemas and SQL queries using SQLAlchemy ORM

- Achieved 95%+ test coverage by developing comprehensive Pytest suites covering edge cases and defined acceptance criteria
- Reduced long-term maintenance overhead by closing 8+ workflow gaps through modular architecture and clean code practices

**MAGIC: THE GATHERING CARD SCANNER** | Python, Flask, OpenCV, OCR, SQLAlchemy | [Github Repo](#)

- Built a Flask-based image-processing API with a robust OCR pipeline for semi-structured data extraction, increasing card recognition accuracy by 22%.
- Documented all API endpoints, extraction logic, and pipeline workflows; leveraged AI-assisted tooling to accelerate development while maintaining full ownership of what shipped.

## EDUCATION

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**SENECA POLYTECHNIC**

Bachelor of Engineering - Software Engineering

North York, ON

September 2023 - Present

**Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Software Development Lifecycle (SDLC), Database Management Systems, Web Development, Operating Systems, Agile Software Development, Artificial Intelligence, Data Mining, Machine Learning